

13 Risks and Outlook

We already addressed this question in the foreword: is the glass half full or half empty? AI tools undoubtedly help to increase your efficiency as a developer or administrator, as a programmer, or as a database designer. However, it's equally undisputed that AI tools create errors, sometimes even major ones.

The errors are all the more serious because current AI tools lack inherent self-control or self-criticism. Instead of a simple “I don't know,” you'll receive a polite answer that is elegantly formulated and convincingly argued, but nevertheless completely wrong.

In this chapter, we take a look at the risks and problems of AI tools:

- First, we provide an overview of the basic risks: outdated code, poor maintainability, data protection concerns, and so on.
- In [Section 13.2](#), we provide concrete examples in which AI tools have returned clearly incorrect answers. Many examples have been taken from our own work. We also cite some studies on this topic.
- Is it even ethically justifiable to use AI tools? In our view, there is no objectively clear answer to this question. However, we're trying to explore the ethical and, in some cases, legal aspects of using AI tools for programming.
- Finally, we summarize where AI-supported coding currently stands and where we think the journey is heading. To get one point out of the way, we're convinced that the use of AI tools is only responsible within the scope of one's own knowledge. AI tools are therefore no substitute for solid IT training.

13.1 Issues and Limitations of Using AI Tools

While working on this book and while programming for other projects, we encountered various issues when using AI coding aids. Accordingly, we've repeatedly referred to limitations in the preceding chapters. At this point, we want to summarize once again where we believe the biggest limitations and pitfalls can be found when using AI tools.

13.1.1 Old, Cumbersome Code

Any language model is only as good as its training material. For software development in particular, training suffers from the fact that there is always more old material available on the internet than up-to-date material. There is an infinite number of pages that refer to old Java versions, but only very few pages in which new functions added to Java are used in a way that makes sense. (You can replace Java with the programming language of your choice in this and the following paragraphs.)

ChatGPT therefore tends to suggest code that uses old, established functions or libraries. That code will usually work. However, if you use a current version of Java, there might have been a new, more elegant, and more efficient way to solve your problem. You can try to specify the version number you use in the prompt (say: Java 21), but even that only helps to a limited extent.

13.1.2 The Question of Maintainability

AI tools enable you to produce lots of lines of code in no time. Your productivity measured in this way will be great, and your boss will be happy about you. But who maintains this code? Who fixes a bug that only occurs under very specific circumstances and is discovered for the first time after three months?

We've already shown in this book that ChatGPT and others can also be a great help in troubleshooting. But what is even better is compact, well-designed code that works right from the start.

Note that by no means do we deny that AI tools often provide excellent code. Unfortunately, this isn't always the case. An experienced developer will probably scrutinize the AI code and perhaps make a second attempt. But many programmers will be pragmatic and carry out a short test to see whether the new function does what it's supposed to do and leave it at that—even if, objectively speaking, the code isn't ideal.

13.1.3 Privacy

If you use free AI services from commercial providers (OpenAI, Microsoft, Google, etc.), your prompts, the resulting responses, and your reaction to them will be stored and used to train future language models. So far, so fair. As with many other services that are free, you pay with your data.

For many companies, however, code is a company secret. Self-developed algorithms make up part of the company's value. Such companies don't feel comfortable giving editors with AI support virtually unhindered access to the entire code.

Of course, AI company offerings (e.g., GitHub Copilot Business) promise a certain level of privacy. However, it's difficult to assess, let alone verify, how far this goes and whether the rather vague promises ("your data is excluded from training by default") are actually kept.

Even if AI companies try to comply with privacy regulations to the best of their knowledge and belief, mistakes do happen. AI tools are slapped together; whoever is first on the market with good AI products can gain significant market share. Quality control and code reviewing can sometimes fall by the wayside.

Moreover, since Edward Snowden's revelations, it's not a conspiracy theory but a fact that the US intelligence services have almost unrestricted access to US cloud storage. AI companies that operate almost exclusively in the United States are hardly exempt from this.

Do you store your code in GitHub repositories? Do you use Visual Studio Code (VS Code) with all conceivable plug-ins? Then, your code is already doubly compromised. The use of a commercial coding wizard doesn't dramatically worsen the situation.

Conversely, if you're reluctant to use AI tools for coding due to privacy considerations, you also need to question your other security measures. You should definitely run your Git repositories yourself (e.g., in a GitLab instance on a company server). You should also be very selective when choosing development tools and the associated plug-ins. The editor must be able to read your code, as must most plug-ins. A malicious plug-in can therefore compromise your code.

13.1.4 A New Kind of Dependency

We don't know how often or how actively you cycle. You have the choice of relying solely on your own muscle power or being assisted by a motor. The first ride on an e-bike is definitely a pleasure! Regardless of whether you want to conquer a small hill or an entire mountain pass—you'll be much faster and feel more comfortable.

It doesn't take long for you to get used to the support provided by the motor and battery. "Ordinary" cycling is no longer fun at first, and a year or two later, you'll also lack muscle power. There's no turning back, and the temptation to use turbo mode (or whatever the highest support level on your e-bike is called) more and more often becomes ever greater.

You can probably guess what this comparison is aimed at. Programming with AI support isn't only more efficient, it's also more convenient. Over time, you no longer know the syntax of frequent constructions by heart, no longer think when formulating loops, and begin to rely more and more on the code provided by your AI tool. You'll run a quick test to see if everything works as expected and then it's on to the next step!

It doesn't take long before you become dependent on AI tools. Even if you haven't (completely) forgotten how to program "properly," without AI sup-

port you won't be as agile and efficient as you used to be.

This can be particularly fatal if political, economic, or legal reasons mean that AI services can no longer be offered in a country. Imagine a court ruling that AI services from OpenAI, Microsoft, Apple, or another IT company don't comply with EU data protection regulations. To avoid a long dispute or high payments, the service will simply get switched off in the EU. That would be a rather unpleasant prospect, also for competitive reasons. (How are European developers without AI support supposed to keep up with American programmers using these tools?)

13.1.5 Learning to Program in the Age of AI

Basically, this book is intended for readers who already know how to program. But let's take a very brief look at the next generation. Learning new skills requires that you actually practice them. Writing essays is certainly a tedious exercise at school, but it's essential if you want to be able to formulate convincing texts yourself later on. Similarly, the initially frustrating compilation of your first lines of code is important for learning basic programming techniques.

Finding a responsible scale for the use of AI tools is currently a huge challenge for schools and universities. To make matters worse, there is a lack of suitable control mechanisms. Self-programmed code is best recognized by the errors it contains. But it's of course inadmissible to immediately place every usable code under general suspicion: "But you did this exercise with AI help, didn't you?"